

System Design Document (SDD)

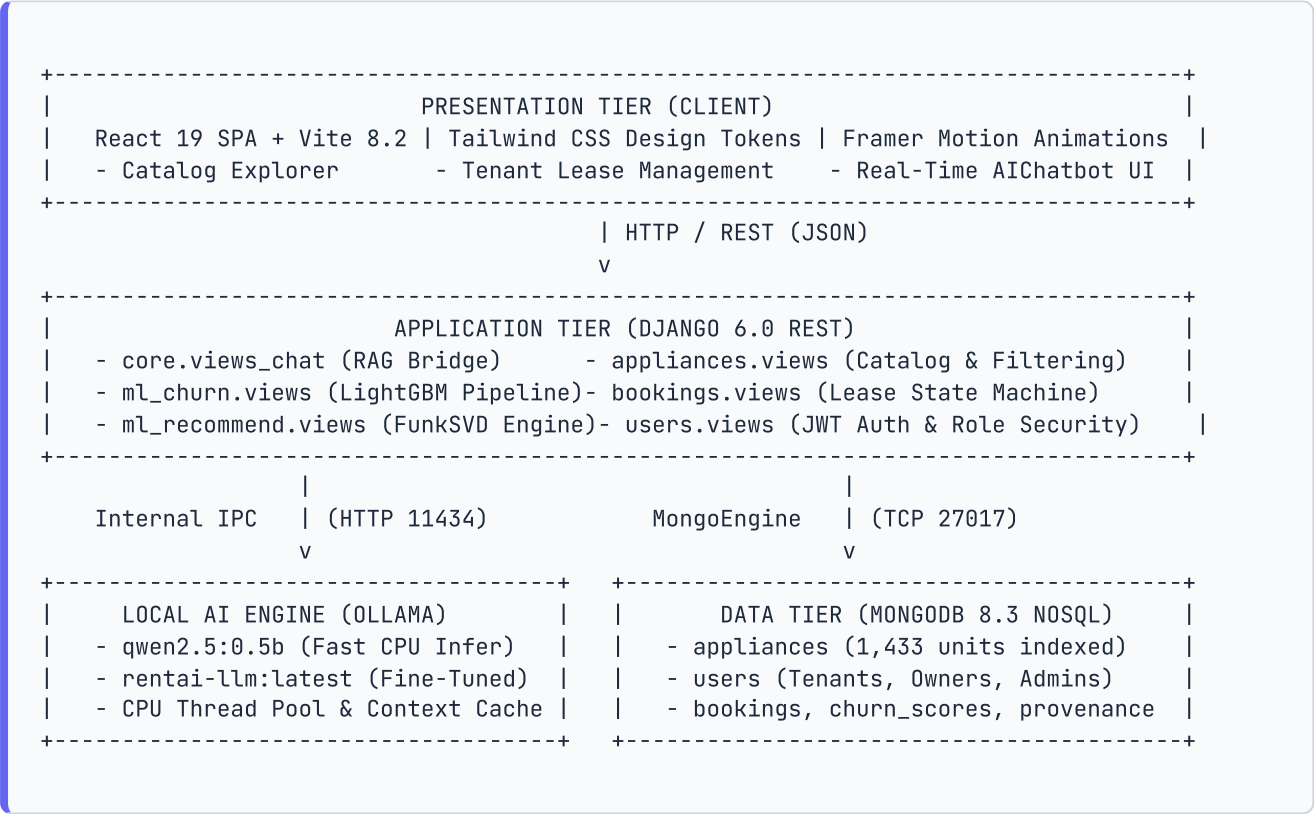
High-Level System Architecture, Tier Decomposition & Distributed Data Flow

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

Document ID:	SDD-SDC2-2026-V2.4	Version:	2.4.0 (Production)
Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Architectural Style & Tier Decomposition

RentAI employs a clean 3-tier decoupled microservice-ready architecture. The presentation layer communicates with the backend via stateless RESTful JSON contracts, backed by a high-throughput NoSQL document repository and isolated machine learning inference engines.



2. Component Integration & Responsibilities

2.1 Authentication & Security Gateway

Managed via SimpleJWT. Every protected endpoint inspects the `Authorization: Bearer <token>` header. Role-Based Access Control (RBAC) middleware isolates tenant actions from host appliance modifications and administrative overrides.

2.2 Automated Ollama Daemon Controller

The backend features an autonomous supervisor function: `auto_start_ollama_if_needed()`. If port 11434 is inaccessible, Django spawns `ollama serve` as a background detached process (using Windows `CREATE_NO_WINDOW`), polling until port readiness before dispatching inference jobs.